

Exercise

Vector Space

(1) $\{v_1, v_2, \dots, v_n\}$ is linearly independent if and only if

$$\forall a_i \in \mathbb{R}, 1 \leq i \leq n,$$

$$\sum_{i=1}^n a_i v_i = 0$$

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$$\text{we have } a_1 = a_2 = \dots = a_n = 0$$

$$\Leftrightarrow \begin{pmatrix} | & | & | & \dots & | \\ v_1 & v_2 & v_3 & \dots & v_n \\ | & | & | & \dots & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = 0 \quad \text{has one only trivial solution.}$$

$$\Leftrightarrow \begin{pmatrix} | & | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & | & \dots & | \end{pmatrix} \text{ is invertible}$$

One can also use dimension to argue :

$$\star \Leftrightarrow \text{denote } A := \begin{pmatrix} | & | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & | & \dots & | \end{pmatrix}, \dim(\text{Col } A) \leq n$$

Since $\{v_i\}_{i=1}^n$ are linearly independent, we have $\dim(\text{Col } A) \geq n$

i.e. $\dim(\text{Col } A) = n$, i.e. $\{v_1, v_2, \dots, v_n\}$ is a basis of

$\mathbb{R}^n$, By Cor 6.1, it is equivalent to $\det(A) \neq 0$.

(2) Let the matrix be A.

□

$$(a) \det A = \det \begin{pmatrix} 2 & 0 & 3 \\ 1 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix} + \det \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$$= 2 \det \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} + 3 \det \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} + (-1)(-1) \overset{+3}{\det \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}} + \det \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$= 11 + (-9)$$

$$= 2$$

(b) $\det A = 2 \neq 0 \Leftrightarrow$ Row vectors are linearly independent.

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(3), let $v = (a, b, c)$. $a, b, c \in \mathbb{R}$.

$$\begin{aligned} \{ (a, b, c) \in \mathbb{R}^3 : \det \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & c \end{pmatrix} \neq 0 \} \\ = \{ (a, b, c) \in \mathbb{R}^3 : c \neq 0, a \in \mathbb{R}, b \in \mathbb{R} \} \end{aligned}$$

(4) Compute for $\det \begin{pmatrix} 1 & 1 & a \\ 0 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \neq 0$

$$\begin{aligned} \text{LHS} &= \det \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} + (-1) \det \begin{pmatrix} 1 & a \\ 1 & 3 \end{pmatrix} \\ &= 1 - (3 - a) \neq 0 \end{aligned} \Rightarrow a \neq 2$$

(b) Follow the hint

$\{v_1, v_2\}$ is a basis iff

$$A = \begin{pmatrix} v_1 & v_2 \end{pmatrix} \text{ is invertible.}$$

One checks that

$$\begin{pmatrix} v_1 & v_2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} v_1+v_2 & v_1-v_2 \end{pmatrix} := A'$$

Hence equivalently $\overset{\text{invertible}}{A'}$ is invertible.

(By (1),
 or Corollary) $\{v_1+v_2, v_1-v_2\}$ is invertible

□

Linear Transformation

(a) Yes. • $T(x_1+y_1, x_2+y_2, x_3+y_3) = (x_1+y_1, x_2+y_2)$
 $= (x_1, x_2) + (y_1, y_2)$
 $= T(x_1, x_2, x_3) + T(y_1, y_2, y_3)$
• $T(cx_1, cx_2, cx_3) = (cx_1, cx_2) = c(x_1, x_2)$
 $= cT(x_1, x_2, x_3)$

where $x_i, y_i, c \in \mathbb{R}$.

(b) No. Not a well-defined map. since x_3 is free.

(c) Yes. For any $A, B \in M_2$, and $c \in \mathbb{R}$.

One checks:

- $T(A+B) = (A+B)^T = A^T + B^T = T(A) + T(B)$
- $T(cA) = (cA)^T = cA^T = cT(A)$

(d) No. $T(0) = 0 + I_2 = I_2 \neq 0$, violating Thm 7.4

(e) Yes. $\forall f, g \in V, c \in \mathbb{R}$.

$$T(f+g) = (f+g)(1) = f(1) + g(1) \\ = T(f) + T(g)$$

$$T(cf) = (cf)(1) = cf(1) = cT(f)$$

$$(2) (a) R(T) = \mathbb{R}^2, \quad N(T) = \{(0, 0, c) : c \in \mathbb{R}\}$$

$$(c) R(T) = M_2, \quad N(T) = \{0\}$$

$$(e) R(T) = \mathbb{R}, \quad N(T) = \{f \in V : f(0) = 0\}$$

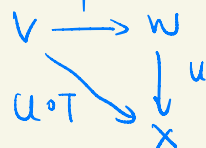
(3) Easy to verify that

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ forms a basis.}$$

T is fixed once we assign a value for

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{pmatrix} a \\ b \end{pmatrix}, \quad a, b \in \mathbb{R}$$

$$4) \text{rank}(U \circ T) \leq \text{rank}(T),$$



$$\text{Let } U|_{R(T)} : R(T) \rightarrow X$$

be the linear transformation that restricts U on the subspace $R(T)$.

This is indeed a linear transformation. Check!

$$\text{Note that } R(U \circ T) = R(U|_{R(T)}) \quad \star$$

This is b/c. $\forall w \in R(U \circ T), \exists v \in V$ s.t.

$$w = U \circ T(v) = U|_{R(T)}(T(v)) \in R(U|_{R(T)})$$

Conversely. $\forall w \in R(U|_{R(T)}), \exists u \in R(T)$

$$\text{s.t. } w = U|_{R(T)}(u) \text{ and } \exists v \in V$$

$$\text{s.t. } u = T(v)$$

$$\text{We have } w = U|_{R(T)} \circ T(v) = U \circ T(v) \in R(U \circ T)$$

$$\text{We have } \text{rank}(U \circ T) \stackrel{\text{def}}{=} \dim(R(U|_{R(T)})) \stackrel{\text{Cor. 7.25}}{\leq} \dim(R(T)) = \text{rank}(T)$$

By Dimension Formula.

$$\text{nullity}(U \circ T) = \dim V - \text{rank}(U \circ T) \quad (1)$$

$$\text{nullity}(T) = \dim V - \text{rank}(T) \quad (2)$$

(1), (2) and the previous result, we done //

(b) True. It suffices to show

$\text{rank}(U \circ T) \geq \text{rank}(T)$ for invertible U .

Find a basis of $R(T)$, say

$$\{v_1, v_2, \dots, v_n\}$$

U invertible. $\Rightarrow \{U(v_1), U(v_2), \dots, U(v_n)\}$ is a

linearly independent spanning set, i.e. a basis for

$$R(U \circ T)$$

Hence $\dim(R(U \circ T)) = \dim(R(T))$, i.e. $\text{rank}(U \circ T) = \text{rank } T$.

Second result follows by **Dimension Theorem**.

□

(b) True.

By previous result, $R(U \circ T) = R(U|_{T(V)})$

is a linear subspace of $R(U)$

Hence. $\dim R(U|_{T(V)}) \leq \dim R(U)$

i.e. $\text{rank}(U \circ T) \leq \text{rank } U$

(c) False. Let $\mathbb{R}^2 \xrightarrow{T} \mathbb{R}^3$



where $T(x_1, x_2) = (x_1, x_2, 0)$

$$U(x_1, x_2, x_3) = 0$$

$$\text{nullity}(U \circ T) = \dim(\mathbb{R}^2) = 2 < 3 = \dim \mathbb{R}^3 = \text{nullity}(U)$$

→ ←

$$(6) \quad T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

Typo ($T: \mathbb{R}^4 \rightarrow \mathbb{R}^4$)

$$R(T) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

$$\Rightarrow \text{rank } T = \dim R(T) = 2$$

$$\text{nullity } T = \dim \mathbb{R}^4 - \text{rank } T = 4 - 2 = 2 \neq 0.$$

Cor 7.33 $\Rightarrow$ T is not injective.

(7) Check for $y_i, x_i, c \in \mathbb{R}$

$$\bullet T'((x_1+y_1, x_2+y_2, x_3+y_3, x_4+y_4, x_5+y_5, x_6+y_6))$$

$$\stackrel{\text{def}}{=} (T(x_1+y_1, x_2+y_2, x_3+y_3), T(x_4+y_4, x_5+y_5, x_6+y_6))$$

$$\stackrel{\text{linear}}{=} (T(x_1, x_2, y_3) + T(y_1, x_2, y_3), T(x_4, x_5, y_6) + T(y_4, y_5, y_6))$$

$$= (T(x_1, x_2, y_3), T(x_4, x_5, y_6)) + (T(y_1, x_2, y_3), T(y_4, y_5, y_6))$$

$$= T'(x_1, \dots, x_6) + T'(y_1, \dots, y_6) \quad \checkmark$$

$$\bullet T'(cx_1, cx_2, \dots, cx_6) = (T(cx_1, \dots, cx_3), T(cx_4, \dots, cx_6))$$

$$= (c T(x_1, x_2, x_3), c T(x_4, x_5, x_6))$$

$$= c (T(x_1, x_2, x_3), T(x_4, x_5, x_6))$$

$$= c T'(x_1, x_2, x_3, \dots, x_6) \quad \checkmark$$

(8) • Verify this is a linear map

• Note that $T(c) = c' = 0$ for $c \neq 0$ in $\mathbb{R}$,

Hence $N(T)$ is not trivial, i.e. T is not injective